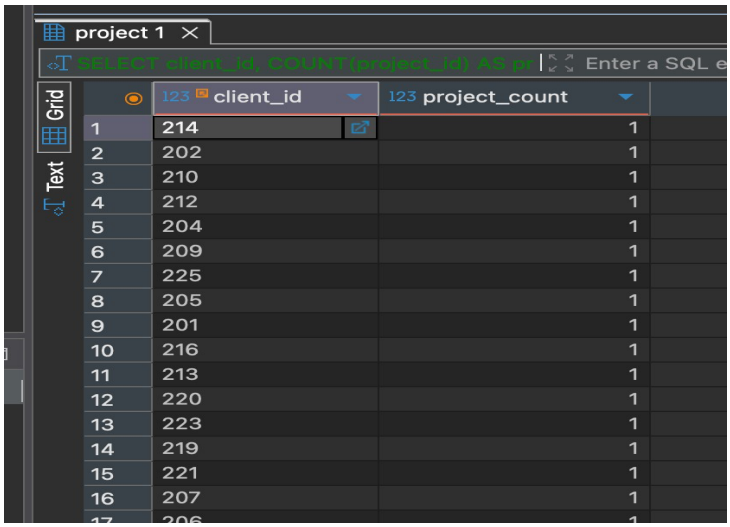
# TASK 5 — AGGREGATION

## Q1) Find the number of projects per client.

**Query:**

```
SELECT client_id,
      COUNT(project_id) AS project_count
FROM project
GROUP BY client_id;
```

**Explanation:**
This query calculates the total number of projects associated with each client. The records are grouped by client_id, and the COUNT function is used to determine how many projects each client has.



| | client_id | project_count |
|---|---|---|
| 1 | 214 | 1 |
| 2 | 202 | 1 |
| 3 | 210 | 1 |
| 4 | 212 | 1 |
| 5 | 204 | 1 |
| 6 | 209 | 1 |
| 7 | 225 | 1 |
| 8 | 205 | 1 |
| 9 | 201 | 1 |
| 10 | 216 | 1 |
| 11 | 213 | 1 |
| 12 | 220 | 1 |
| 13 | 223 | 1 |
| 14 | 219 | 1 |
| 15 | 221 | 1 |
| 16 | 207 | 1 |

## Q2) Find the number of employees per department.

**Query:**

```
SELECT dept_id,
      COUNT(employee_id) AS total_employees
FROM employee
GROUP BY dept_id;
```

**Explanation:**
This query counts the number of employees in each department. It groups the data by dept_id and uses the COUNT function to return the total number of employees per department.

| | dept_id | total_employees |
|---|---|---|
| 1 | 103 | 4 |
| 2 | 101 | 4 |
| 3 | 116 | 4 |
| 4 | 122 | 4 |
| 5 | 121 | 4 |
| 6 | 117 | 4 |

## Q3) List employees along with their department names.

**Query:**

```
SELECT e.employee_fname,
       e.employee_lname,
       d.dept_name
FROM employee e
JOIN department d
ON e.dept_id = d.dept_id;
```

**Explanation:**
This query retrieves employee first and last names together with their corresponding department names. It uses an inner join between the employee and department tables based on the dept_id column.

| | employee_fname | employee_lname | dept_name |
|---|---|---|---|
| 1 | Zara | Ali | Human Resources |
| 2 | Aarav | Shukla | Human Resources |
| 3 | Varun | Khanna | Human Resources |
| 4 | Amit | Sharma | Human Resources |
| 5 | Adarsh | Nigam | Finance |
| 6 | Bhavya | Mittal | Finance |
| 7 | Sneha | Joshi | Finance |
| 8 | Riya | Patel | Finance |
| 9 | Binit | Choudhary | Sales |
| 10 | Chirag | Bhardwaj | Sales |
| 11 | Abhishek | Tiwari | Sales |
| 12 | Rahul | Verma | Sales |

## Q4) Find the number of employees working per project.

**Query:**

```
SELECT p.project_name,
      COUNT(e.employee_id) AS employee_count
FROM project p
JOIN team t
ON p.project_id = t.project_id
JOIN employee e
ON t.team_id = e.team_id
GROUP BY p.project_name;
```

**Explanation:**
This query calculates the number of employees assigned to each project. It joins the project, team, and employee tables and groups the results by project name, using the COUNT function to compute the employee total per project.

| | project_name | employee_count |
|---|---|---|
| 1 | E-commerce Portal | 4 |
| 2 | Inventory System | 4 |
| 3 | Payroll System | 4 |
| 4 | Banking App | 4 |
| 5 | Cloud Migration | 4 |
| 6 | Analytics Dashboard | 4 |
| 7 | IoT Monitoring | 4 |
| 8 | CRM Upgrade | 4 |
| 9 | ERP Deployment | 4 |
| 10 | Website Redesign | 4 |
| 11 | Smart Parking | 4 |
| 12 | Networking Upgrade | 4 |
| 13 | HR Automation | 4 |
| 14 | Telemedicine Platform | 4 |
| 15 | Video Streaming App | 4 |
| 16 | Document Management | 4 |